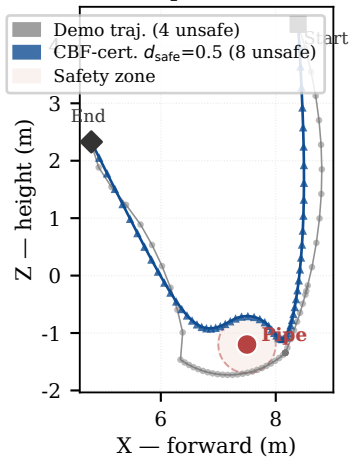
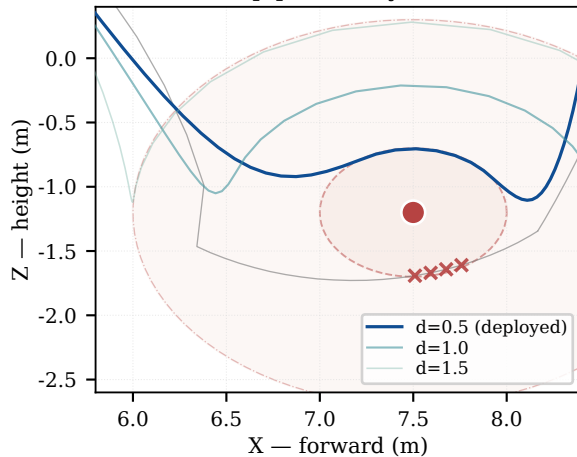


CBF Trajectory Safety Certification — Pipeline Avoidance ($d_{\text{safe}}=0.5$ m, $\lambda=0.3$)

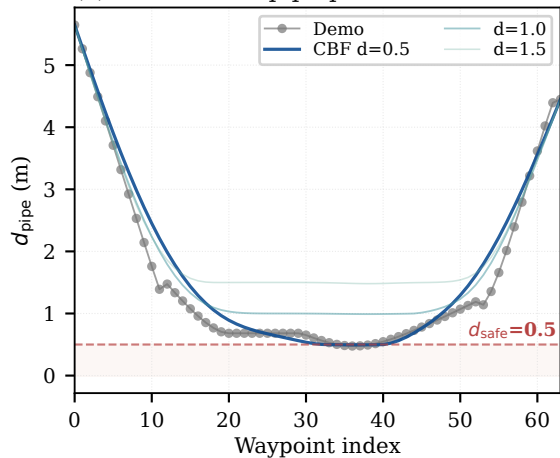
(a) XZ-plane overview ($d_{\text{safe}}=0.5$ m)



(b) Zoom — pipe vicinity



(c) Distance-to-pipe profile



(d) Distance distribution — before vs after

